χ: Symmetry Understanding of 3D Shapes via Chirality Disentanglement

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Motivation

Research question: Can we leverage image foundation models to understand symmetries of 3D shapes?
Motivation: Generally, 3D shape feature extractors are often **limited** in their ability to **understand chirality**, as seen when using Diff3F [1] features to match two shapes:

Source | Diff3F [1] | Diff3F [1] features 2-sets clustering

Terminology:

- **Chirality:** Symmetry and chirality are two key concepts of geometry understanding. While **symmetry** highlights the **similarities** between two parts, **chirality** emphasizes their **differences**.
- **Foundation model:** A **large-scale machine learning model trained on broad data** that can be adapted to perform a **wide range of tasks** across different domains.

Contributions

A method that **distinguishes left and right** of shapes.

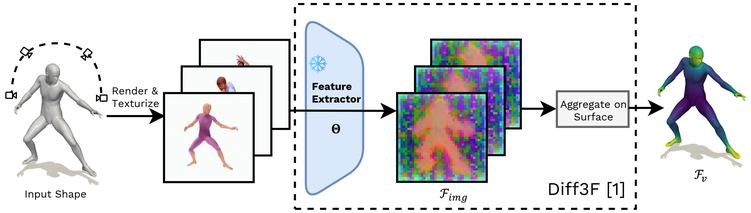
- We achieve this by **extracting** a **chirality feature** out of given shape features in an **unsupervised** manner.
- We demonstrate that our method **outperforms state-of-the-art** methods in **left-right distinguishment**.

Our method effectively solves left-right disambiguity in downstream tasks.

- We show that in **shape matching**, concatenating our feature with Diff3F [1] features significantly **reduces chirality errors**.

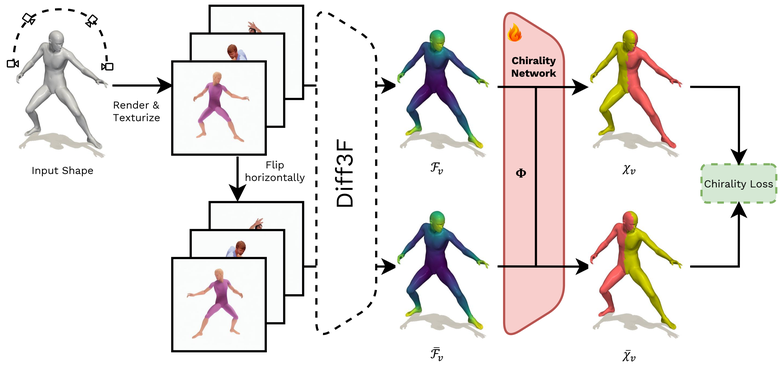
Related works

- **Symmetry detection methods**: Existing methods are **mainly handcrafted**. Zhang et al. [4] rely on a signed angle-based feature, while Liu et al. [3] identify extremal points via average geodesic distance to find symmetry maps.
- **3D Shape features**: There is a **variety** of **learned and handcrafted** vertex features in 3D shape understanding. One **recent** example is **Diff3F** [1], which leverages **foundation image models** to decorate shape vertices with semantic information.



Method

→ **Pipeline**



For a given shape $\mathcal{X} = (V, E)$

→ **Chiral pair generation**: For each vertex of the shape $v \in V$, its feature $\mathcal{F}_v$ and its semantic-flipped feature $\bar{\mathcal{F}}_v$ are generated using rendered views and **flipped** rendered views, respectively.

→ **Chirality network**: The chirality network consists of an encoder g_Φ, a decoder h_Ψ and a projection matrix A.

→ **Chirality feature extraction**: per-vertex chirality feature $\chi_v = \frac{[Ag(\mathcal{F}_v)]_1}{\|Ag(\mathcal{F}_v)\|_2}$.

Concatenating χ_v of all vertices to form a chirality feature vector χ, and $\bar{\chi}$ is given by the chirality network using $\{\bar{\mathcal{F}}_v\}_{v \in \mathcal{X}}$.
Concatenating $\mathcal{F}_v$, $\bar{\mathcal{F}}_v$ of all vertices to form feature vectors $\mathcal{F}$ and $\bar{\mathcal{F}}$ respectively.

→ **Losses**

- Dissimilarity loss: To **maximise the difference** between $\mathcal{F}_v$ and $\bar{\mathcal{F}}_v$

$$\mathcal{L}_{\text{dis}} = -\frac{1}{|V|}\|\chi - \bar{\chi}\|_2$$

- Total variation loss: To achieve **spatial smoothness**

$$\mathcal{L}_{\text{var}} = \frac{1}{|E|}\sum_{(u,v)\in E}\|\chi_u - \chi_v\|_1 + \|\bar{\chi}_u - \bar{\chi}_v\|_1$$

- Invertibility loss: To keep our encoder g from learning **trivial solutions**

$$\mathcal{L}_{\text{inv}} = \frac{1}{\sqrt{|V|}}\|[\mathcal{F}^T \;\; \bar{\mathcal{F}}^T]^T - h(g([\mathcal{F}^T \;\; \bar{\mathcal{F}}^T]^T))\|_F$$

- Fifty-fifty loss: To make both halves have the **same** number of assigned vertices

$$\mathcal{L}_{\text{fif}} = \frac{1}{|V|}\left(\frac{\chi^T \mathbf{1}_{|V|}}{\|\chi\|_\infty} + \frac{\bar{\chi}^T \mathbf{1}_{|V|}}{\|\bar{\chi}\|_\infty}\right)$$

Results

→ **Left/Right distinguishment**

Our method significantly outperforms its competitors quantitatively and qualitatively

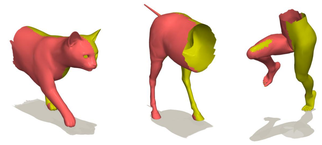
Train	BeCoS	BeCoS-h		BeCoS-a		FAUST		SMAL	
Test	BeCoS	BeCoS-h	BeCoS-a	BeCoS-h	BeCoS-a	FAUST	SCAPE	SMAL	TOSCA
Liu et al. [3]	79.98	79.83	80.46	79.83	80.46	90.45	80.84	75.71	72.88
Ours	91.84	94.09	84.19	90.36	91.10	94.76	95.51	96.59	94.09

→ **Left/Right distinguishment generalisation**

Training on complete isotropic shapes, and test on:

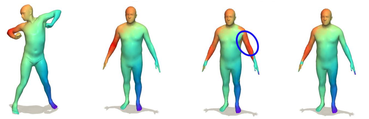
- **Anisotropic shapes**

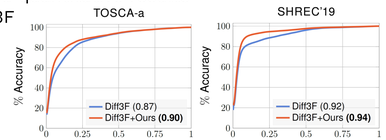
- **Partial shapes**



→ **Shape matching**

When combined with Diff3F features [1], our method effectively resolves left/right ambiguity, as seen in both the qualitative and quantitative results





Reference

1. Dutt et al. Diffusion 3D Features (Diff3F): Decorating Untextured Shapes with Distilled Semantic Features. CVPR 2024.
2. Zhang et al. Telling Left from Right: Identifying Geometry-Aware Semantic Correspondence. CVPR 2024.
3. Liu et al. Finding Surface Correspondences Using Symmetry Axis Curves. Computer Graphics Forum, 2012.
4. Zhang et al. Symmetry Robust Descriptor for Non-Rigid Surface Matching. Computer Graphics Forum, 2013

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